

Do Mandatory Risk Factor Disclosures Reduce Stock Price Crash Risk?

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Abstract

We test whether voluntary or mandatory risk factor disclosures (RFDs) in 10-K filings is associated with a reduction in stock price crash risk. We find that the level of mandatory RFDs in 10-K filings is associated with a reduction in stock price crash risk, but find no similar relationship for voluntary RFDs. We exploit two exogenous shocks to mitigate endogeneity concerns that remain to be addressed in the literature. We investigate the moderators for this relationship and find the reduction is magnified among firms with higher information asymmetry, higher litigation risk, or better corporate governance. Overall, our findings identify a potential avenue to mitigate stock price crash risk and provide evidence that mandated RFDs contain useful information content and benefit investors.

Keywords: Item 1A, Risk factor disclosures, Crash risk, Information asymmetry, Corporate governance, Litigation risk.

JEL Classification: G10

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Abstract

We test whether voluntary or mandatory risk factor disclosures (RFDs) in 10-K filings is associated with a reduction in stock price crash risk. We find that the level of mandatory RFDs in 10-K filings is associated with a reduction in stock price crash risk, but find no similar relationship for voluntary RFDs. We exploit two exogenous shocks to mitigate the endogeneity concerns that remain to be addressed in the literature. We investigate the moderators for this relationship and find the reduction is magnified among firms with higher information asymmetry, higher litigation risk, or better corporate governance. Overall, our findings identify a potential avenue to mitigate stock price crash risk and provide evidence that mandated RFDs contain useful information content and benefit investors.

1 Introduction

In 2005, the Securities and Exchange Commission (SEC) mandated risk factor disclosures¹ (RFDs) in 10-K filings. Subsequent research has found these disclosures affect firms' information environment, finding effects in stock return volatility, analysts' forecast dispersion, bid-ask spreads, leverage, investment, and credit default swap (CDS) spreads.² However, by employing text-analysis software to read through entire 10-K filings using the Campbell et al. (2014) dictionary, we find that even before 2005, many firms disclose risks voluntarily in their management discussion or in risk-disclosure sections of their filings. Our paper uses these findings to investigate whether the SEC's mandate of RFDs did indeed affect firms' information environment. We examine RFDs' impact on an important proxy for the quality of the information environment, stock price crash risk—the chance a firm will experience a large decline in stock returns.

Using a sample of US public firms from 1996 to 2013, we investigate three questions: (1) Does the SEC's mandate of RFDs enhance firms' information environment and consequently reduce firms' stock price crash risk? (2) Do voluntary RFDs also affect stock price crash risk? (3) What is the moderator for this effect: improved information asymmetry, shareholder power (litigation and governance effectiveness), or both?

Our results show that RFDs reduce stock price crash risk, as measured by two commonly used proxies in the literature: *NCSKEW*, whether the distribution of returns has a negative skew, and *DUVOL*, the down-to-up volatility or the ratio of the standard deviation of the past down (negative return) weeks to the standard deviation of the up weeks. However, this effect only

¹ We use risk factor disclosure and risk disclosure interchangeably hereafter.

² Kravet and Muslu 2013; Campbell et al. 2014; Hope et al. 2016; Avramov et al. 2018; and Chiu et al. 2018.

appears in the post-2005 period, when the SEC mandated risk factor disclosures; before this period, there was no statistically significant relationship between RFDs and stock price crash risk. Therefore, it appears the SEC's mandate improves the information environment around the firm; voluntary disclosure of RFDs has no impact on the information environment.

Further, we investigate information asymmetry and shareholder power as mechanisms for this reduction in stock price crash risk. First, the theoretical frameworks by Hong and Stein (2003), Jin and Myers (2006), and others (e.g., Kothari et al. 2009; Bhargava et al. 2017) show that information asymmetry between corporate insiders and investors causes stock price crashes. Kravet and Muslu (2013) and Campbell et al. (2014) suggest that enhanced RFDs reduce information asymmetry and allow investors to account for unfavorable firm-specific information. Consequently, we expect the unfavorable information to be impounded into firms' stock prices on a regular basis, mitigating the possibility that bad-news hoarding leads to a stock price crash.

We propose and test the hypothesis that the enhanced risk disclosure plays a more critical role when the information environment is less transparent. Our results show that risk disclosure has a stronger effect on mitigating stock price crashes when a firm has higher information asymmetry, such as when the firm's underlying fundamentals are more volatile, when the firm's business operations are more complex, or when the market is more uncertain about the firms. These findings provide empirical evidence that RFDs matter more for firms with higher information asymmetry.

Second, the effect of RFDs on crash risk can also work through shareholder power, defined as the ability of investors to influence firm management. Previous research finds that the lack of shareholder power is linked to stock price crash risk: poorly governed firms could restrict information that could adversely impact stock performance and keep it from being impounded into

firms' stock price (Baginski et al. 2002; An and Zhang 2013). And Bhargava et al. (2017) show evidence that shareholder protection is associated with stock price crash risk. We measure shareholder power using two proxies: litigation and governance. Litigation is one means for shareholders to enforce their rights. We conjecture that litigation risk motivates managers to reveal bad news with lesser delay (Baginski et al. 2002), which in turn reduces stock price crash risk. We observe a stronger relation between RFDs and crash risk for firms with higher litigation risk as measured by Kim and Skinner (2012). We also find that, conversely, the passage of universal demand (UD) laws makes shareholder litigation more difficult (Nguyen et al. 2018) and is associated with a weaker relationship between RFDs and crash risk. Consequently, RFDs complement litigation in reducing stock price crash risk.

Additionally, we expect good governance mechanisms to play a role in moderating the effect of RFDs on crash risk. Well-governed firms will release information in a more timely and relevant manner and are less likely to engage in hoarding bad news (Beekes and Brown 2006; Cerbioni and Parbonetti 2007; Brown et al. 2011). Our results show that RFDs induce an outsized reduction in crash risk for firms with better governance, as measured by the entrenchment index (E-index) and institutional ownership, suggesting that good governance reinforces the quality of information disclosure. Additionally, we show that the passage of business combination (BC) laws, which typically entrenches management and exacerbates governance, weakens the relationship between RFDs and stock price crash risk.

Our results are robust to endogeneity and a wide variety of other concerns. To mitigate the concern that RFDs and crash risk could be endogenously determined, we rely on two quasi-natural experiments. We conduct a difference-in-differences (DiD) test looking at US vs. Canadian firms. US firms are required to report RFDs after 2005; however, Canadian firms were not subject to this

mandate. We also use a regression discontinuity design (RDD) to examine the impact of the SEC mandating RFDs on all firms with market capitalization over \$75 million but exempting firms below that threshold from the mandate. In both cases, we find that mandating RFDs leads to a decrease in stock price crash risk.

We also conduct a battery of robustness checks, including an alternative measure of crash risk, orthogonalized RFDs through a two-stage regression, and changes in the measures of crash risks on changes in the RFDs. We also control for events like the global financial crisis. Overall, our main results are robust to these alternative specifications, suggesting that our findings are unlikely to be driven by some confounding factors that are not considered in this research design.

This paper contributes to the literature in several ways. First, our findings suggest that mandatory RFDs reduce stock price crash risk. This research therefore adds to the stock price crash risk literature by identifying a potential avenue to mitigate the risk, which is of practical and theoretical interest due to the large impact that stock price crashes have on firm outcomes (Harvey and Siddique 2000; Conrad et al. 2013; Kim and Zhang 2016). In light of our findings, we provide a potential prescription for reducing crash risk: enhancing risk disclosure.

We do recognize that prior literature documents that crash risk is possibly mitigated by greater disclosure and higher transparency (e.g., Hutton et al. 2009; Kim and Zhang 2014; Kim et al. 2016; Kim et al. 2019). However, little has been done to address endogenous concerns. We overcome this limitation by exploiting two exogenous shocks to firm risk disclosures.

Second, this paper adds to the risk disclosure literature. Prior research finds risk disclosures impact stock return volatility, liquidity, CDS spreads, and corporate decisions (Kravet and Muslu 2013; Campbell et al. 2014; Hope et al. 2016; Avramov et al. 2018; Chiu et al. 2018; Au and Tan 2022). For instance, Kravet and Muslu (2013) show evidence that increases in risk disclosures are

associated with higher stock return volatility and trading volume, Avramov et al. (2018) study how text-based downside risk measures impact corporate decisions, and Au and Tan (2019) show that risk disclosure is negatively associated with the subsequent corporate innovation activities. Our paper adds to this strand of literature in two ways. First, we show that RFDs affect stock price crash risk, a proxy for the quality of the firm's information environment. More importantly, we show that it is mandatory, not voluntary, RFDs that impact stock price crash risk.

The remainder of this paper is organized as follows. In Section 2, we review the literature and develop hypotheses. Section 3 describes the data and the construction of our key variables. Section 4 presents our research design and empirical results. Finally, we present our summary and concluding remarks in Section 5.

2 Literature and Hypothesis Development

2.1 Literature Review

Stock price crash risk is defined as the negative skewness in the distribution of returns for an individual stock, or in layman's terms, a higher-than-normal chance of a large negative stock return (Chen et al. 2001; Kim and Zhang 2014). Stock price crash risk is important to investors, especially small investors, and corporate executives because they are generally under-diversified in their portfolios. A large movement in one or two stocks would have deleterious effects on their wealth (Lambert et al. 1991; Barber and Odean 2013). Further, crash risk cannot be diversified away and is a priced factor—investors demand higher returns for higher stock price crash risk—indicating that even general investors care about negative skewness (Harvey and Siddique 2000; Sunder 2010).

The literature differs on whether stock price crash risk is due to information asymmetry or shareholder power. Jin and Myers (2006) show in their theoretical model that investor protections

(α in their model) net out of the model and should have no impact on stock price crash risk. This is further supported by a wide literature that shows that stock price crash risk is related to various aspects of information asymmetry such as earnings management, accounting conservatism, or other informational concerns (Kim et al. 2011; Kim and Zhang 2016; Callen and Fang 2015; Chen et al. 2017). Shareholder power, or the ability of investors to influence firm management, can also underlie the effect of risk disclosures. Bhargava et al. (2017) use takeover protection laws and establish a linkage between bad news hoarding and stock price crash risk. This research is further reinforced by other studies finding that price informativeness is related to investor protection or governance (Fernandes and Ferreira 2009; Ferreira and Laux 2007; Ferreira et al. 2011; An and Zhang 2013; Kim et al. 2016; Xu et al. 2014; Yuan et al. 2016). Our paper shows evidence that both information asymmetry and investor protections are associated with stock price crash risk.

Due to the large impact of crash risk on firm outcomes (Harvey and Siddique 2000; Conrad et al. 2013; Kim and Zhang 2016), recent literature examines the causes of crash risk and potential methods to reduce the risk. The determinants of crash risk identified by the literature include financial reporting opacity (Hutton et al. 2009; Kim and Zhang 2014), tax avoidance (Kim et al. 2011), corporate social responsibility (Kim et al. 2014), accounting conservatism (Kim and Zhang 2016), financial statement comparability (Kim et al. 2016), the deviation of cash flow rights from voting rights (Hong et al., 2017), 10-K report readability (Kim et al. 2019), among others. Habib et al. (2018) provide an excellent survey on the stock price crash risk literature.

A contemporaneous work by Goto et al. (2018) is closely related to but differs from our study. First, Goto et al. (2018) do not address endogeneity and their findings could be potentially subject to either omitted variables or reverse causality. To address this concern in this paper, we employ two quasi-natural experiments of regulatory mandates in 2005 and 2008 by the SEC to

identify the causal effect. Second, while Goto et al. (2018) examines the relationship between market risk disclosure and crash risks, we further investigate which risk disclosure mitigates the subsequent crash risks. Exploring potential mechanisms helps identify how the main effect works in different types of firms or environments. Last, although Goto et al. (2018) document that market risk disclosures are negatively correlated with crash risk, the study does not explore how disclosures of other types of risk, such as systematic, legal, and financial risks, affect crash risk.³ To gauge the overall effect, our study searches for keywords identified in the Campbell et al. (2014) dictionary in the entire 10-K filing whereas Goto et al. (2018) exclusively investigate Item 7A content.

2.2 Hypothesis Development

The Jin and Myers (2006) model notes that stock price crashes are related to information asymmetry. Their model suggests that stock price crashes may occur when managers are unable or unwilling to continue to withhold large amounts of negative information and instead release it all at once. This problem is exacerbated by selective risk disclosure; managers will voluntarily disclose risk information when it is favorable (i.e., low risk) and will withhold information if it is not (i.e., high risk; Jorgensen and Kirschenheiter 2003). Consequently, by mandating additional disclosure in 2005, the SEC requires managers to share risk information regardless of its favorability. Thus, stock price crash risk can be mitigated by mandating adequate disclosure to reduce this information asymmetry.

³ Similarly, our paper looks at all types of disclosed risk while papers like Wang, Kannan, and Ulmer (2013) and Jasa (2020) only look at data/information security risk factors.

Enhanced risk factor disclosure (RFD) should combat these problems and improve the informational environment surrounding the firm, reducing information asymmetry for investors. For example, Kravet and Muslu (2013) show that risk disclosure reveals both known and unknown risk factors, thereby improving investors' risk perception. Campbell et al. (2014) provide evidence that the market's assessment of a firm's risk increases with RFDs and suggest that risk disclosure appears to be firm-specific rather than generic. By investigating the credit market, Chiu et al. (2018) show results that RFDs improve information transparency about the firm's underlying risk and find that the firms with larger information asymmetry benefit more from the risk disclosure. Accordingly, the SEC mandate for risk disclosure should reduce information asymmetry, and thus managers' ability to hoard bad news. In turn, stock price crash risk will be reduced because bad news is impounded by the market on a timelier basis.

On the other hand, we also need to consider the unintended effects of risk disclosure. Huang et al. (2021) find that RFDs have an unintended effect of lowering firms' perceived litigation risk. When the firm's perceived litigation risk is reduced, managers would be more inclined to withhold negative news. Therefore, an increase in the probability of stock price crash would follow. Another issue is that disclosure levels pre-2005 may already have been at optimal levels and increasing disclosure beyond optimal levels would negatively affect a firm's risk level (Jorgensen and Kirschenheiter 2003). Consequently, to find the net effect of risk disclosure on crash risk is an empirical question. This leads to Hypothesis 1:

HYPOTHESIS 1: *Mandatory RFDs are associated with future stock price crash risk.*

In contrast to mandatory RFDs, voluntary disclosure should have limited impact on stock price crash risk. As noted earlier, stock price crash risk is increased by the hoarding of bad news. With voluntary disclosure, managers tend to disclose information that is favorable and withhold

information that is unfavorable (Jorgensen and Kirschenheiter 2003). Thus, voluntary RFDs are predicted not to reduce bad news hoarding or reduce stock price crash risk. This leads to the second hypothesis:

HYPOTHESIS 2: Voluntary RFDs are not associated with future stock price crash risk.

Mandatory risk factor disclosure has been associated with reduced information asymmetry among firms. Evidence shows that mandatory RFDs improve information asymmetry and the firm's information environment, such as trading volume, volatility, etc. (Kravet and Muslu 2013; Hope et al. 2016; Hail et al. 2020). This improved environment should reduce stock price crash risk because it reduces managers' ability to hide bad news from shareholders. Consequently, the effect of mandated risk disclosure on crash risk may be even greater for firms with higher information asymmetry. Therefore:

HYPOTHESIS 3a: The association between mandatory RFDs and stock price crash risk is stronger (i.e., more negative if a negative effect exists) for firms with greater information asymmetry.

We also explore another potential channel for the reduction in stock price crash risk, shareholder power, or shareholders' ability to influence managers. Morck et al. (2000) find that the lack of legal shareholder protections could be linked to informativeness of stock prices because managers at these firms could restrict information to keep it from being impounded into the market. Shareholder litigation can be used as a proxy for shareholder power: if a shareholder has been illegally treated by the firm, such as being denied material information, the shareholder can use litigation to recover economic damages from the firm. The SEC mandates that firms disclose material risk factors that could adversely affect the issuer's business, operations, industry, financial positions, or future performance (IRRC, 2016). Thus, any firm that fails to disclose the meaningful

and firm-specific risk factor information in a timely manner may be subject to possible lawsuits and litigation costs (Robbins and Rothenberg 2005). For example, Skinner (1994) shows that managers have incentives to preempt the announcement of large negative earnings surprises to prevent large stock price declines on earnings announcement dates and thereby reduce the potential legal costs of shareholder lawsuits. Moreover, Brown et al. (2018) further show that peer firms provide more firm-specific disclosures in the year following the SEC comments on the industry leader's risk disclosure. Taken together, a potential mechanism for the impact of RFDs on stock price crash risk could be shareholders' ability to influence managers, such as through litigation. Collectively, these considerations lead to the following hypothesis:

HYPOTHESIS 3b: The association between mandatory RFDs and stock price crash risk is stronger (i.e., more negative if a negative effect exists) for firms with higher litigation risk.

Though shareholder legal rights are relatively consistent across the USA, governance—the ability of shareholders to remove bad managers—can vary dramatically by firm. Firms with better governance typically have more informative disclosure than firms with weaker governance (Beekes and Brown 2006; Cerbioni and Parbonetti 2007; Brown et al. 2011). Hence, if the SEC mandates additional disclosure, managers at well-governed firms are more likely to provide relevant and material information. Conversely, managers in poorly governed firms may continue to hide bad news by disclosing irrelevant or immaterial information. Thus, we develop the following hypothesis:

HYPOTHESIS 3c: The association between mandatory RFDs and stock price crash risk is stronger (i.e., more negative if a negative effect exists) for firms with better governance.

3 Data

Four main data sources were combined for this analysis. The first dataset is the risk disclosures in 10-K filings. We obtained 10-K filings and their risk disclosures from January 1994 to December 2013 from the SEC's Electronic Data Gathering, Analysis, and Retrieval (EDGAR) database.

We use CRSP and Compustat datasets to obtain firm return and characteristics data. The CRSP provides stock return data for the US (AMEX, NASDAQ, and NYSE) markets. We obtain price and trading data from CRSP and annual financial data from Compustat. To be included in the sample, each firm-year observation needs nonnegative total assets (*AT*) and common equity (*CEQ*). All three datasets were matched together using the Central Index Key (CIK) and CUSIP unique company identifiers while also matching by filing dates. Our final sample includes 34,951 firm-year observations from 1996 to 2013. All the risk disclosure and continuous variables are winsorized at the 1st and 99th percentiles each year to reduce the impact of outliers.

3.1 *Measures of Risk Disclosure*

Risk disclosure data is drawn from Au and Tan (2023). The data processing methodology is outlined below. Following Campbell et al. (2014), only 10-K filings with more than 1,000 words are included. Late filings such as NT 10-K, NT 10-KA, and NTN 10-K are excluded to ensure full 10-K filings and not reissuances of the filings are captured. We use text analysis software to parse the risk disclosure sections according to the lexical fields from Campbell et al. (2014). The lexical fields incorporate five risk categories: Financial, Idiosyncratic, Legal and Regulatory, Systematic, and Tax. We combine these risk categories into one aggregate level because investors are interested in all categories of potential risk and not just in one or two categories that a firm may

disclose.⁴ The variable *RFD* is defined as the natural logarithm of the sum of all risk disclosure scores using the Campbell et al. (2014) dictionary⁵ and is similar to the *Allrisk* metric used in Au and Tan (2023). Note that this measure captures the amount of information content (i.e., informativeness), rather than specificity, tone, quality, or readability.

The annual risk disclosure score is calculated as follows. First, for each 10-K, we compute the frequency of the risk words in the entire filing based on the lexical fields for the relevant risks. Next, we scale the frequency of the risk words by the number of words in the 10-K (to account for variability in the length of 10-K) and then multiply this ratio by 100 to express the value as a percentage. If there are multiple 10-Ks for the same firm-year (for example, if there is an addendum), we use the weighted average of the two documents' risk scores weighted by document length.

After 2005, firms were required to include an Item 1A Risk Factors section that discusses the most significant risks to the firm. However, some firms discussed risks in their 10-K filings before 2005, either in a dedicated risk section, in the management discussion, or in notes. We therefore collect 10-K data as well as conduct text analysis before and after 2005 to be able to capture risk disclosures over the entire sample period. The text analysis software combed through the entire 10-K filings to identify RFDs. This means the *RFD* measurement is independent of how risk disclosure is packaged in the 10-Ks and whether Item 1As were mandated by the SEC or not. Thus, risk scores are similar regardless of where in the filing the risks are disclosed.

⁴ This is confirmed by the largely insignificant results (untabulated) of our later baseline regressions of crash risk against individual risk categories.

⁵ The dictionary of words is itemized in Appendix 3 of Campbell et al. (2014).

There is some concern that the 10-K filings may not be valuable information because (1) the text analysis may miscategorize risks, and (2) the risk disclosures may be boilerplate and have no effect on the firm. First, to ensure that the textual analysis results are valid, a manual check was conducted on 500 random sentences the software categorized as having risk. In the manual sample, a human coder confirmed whether the correct risk was identified in each sentence and found that the relevant risk was correctly identified by the software in approximately 85% of the manually checked instances, easily exceeding the 80% standard for textual analysis (Wade et al. 1997; Humphreys and Wang 2018). Second, while risks identified in the 10-K filings may appear to be boilerplate, they have been shown to have impact on important firm characteristics such as information asymmetry, abnormal returns, trading volume, and firm volatility (Campbell et al. 2014; Kravet and Muslu 2013; Hope et al. 2016). Consequently, the text analysis of risk disclosure provides valuable information for this analysis.

3.2 Measures of Stock Price Crash Risk

This paper uses two main metrics for crash risk, negative conditional return skewness (*NCSKEW*) and down-to-up volatility (*DUVOL*) based on CRSP data.⁶ These two measures are based on firm-specific weekly returns, estimated as the residuals from the expanded market model (Kim et al. 2011; Kim et al. 2016; Chang et al. 2017):

$$r_{j,\tau} = \alpha_j + \beta_{1j}r_{m,\tau-2} + \beta_{2j}r_{m,\tau-1} + \beta_{3j}r_{m,\tau} + \beta_{4j}r_{m,\tau+1} + \beta_{5j}r_{m,\tau+2} + \varepsilon_{j\tau} \quad (1)$$

where $r_{j,\tau}$ is the return on stock j in week τ and $r_{m,\tau}$ is the return on the CRSP value-weighted market index in week τ . We include the lead and lag terms for the market index return to allow for

⁶ We also use a third alternative measure *CRASH_DUMMY*, which gauges actual crashes ex post, for robustness checks, and find qualitatively similar results in Section 4.3.1.

nonsynchronous trading (Dimson 1979). The firm-specific weekly return for firm j in week τ , $w_{j,\tau}$, is measured by the natural logarithm of one plus the residual return in Equation (1), that is, $w_{j,\tau} = \ln(1+\varepsilon_{j,\tau})$.

NCSKEW measures the asymmetry of the return distribution by calculating the negative of the third moment of firm-specific weekly returns for each calendar year and normalizing these values by the standard deviation of firm-specific weekly returns raised to the third power. It is calculated using the following formula:

$$NCSKEW_{j,\tau} = -[n(n-1)^{\frac{3}{2}} \sum w_{j,\tau}^3] / [(n-1)(n-2) (\sum w_{j,\tau}^2)^{\frac{3}{2}}] \quad (2)$$

The second metric of crash risk is the down-to-up volatility, *DUVOL*. For firm j over fiscal year τ , firm weekly returns are separated into “down” weeks (when the returns are below the annual mean) and “up” weeks (returns are above annual mean). The standard deviation of each of the down and up weeks is calculated separately and *DUVOL* is the natural logarithm of the ratio of standard deviation of down to up weeks as calculated in equation (3):

$$DUVOL_{j,\tau} = \log\{(n_u - 1) \sum_{Down} w_{j,\tau}^2 / (n_d - 1) \sum_{Up} w_{j,\tau}^2\} \quad (3)$$

3.3 Summary Statistics

Table 1 shows the summary statistics for the final sample of 5,784 unique firms. Panel A shows the full sample statistics. The average value of *NCSKEW* and *DUVOL* is -0.003 and -0.071 , respectively, which is comparable to the figures reported by Chang et al. (2017) over a similar sample period. The distributions of other variables are also largely consistent with those reported in prior studies.

Table 1 Panel B splits the sample into below- and above-median RFD_{t-1} subsamples and then compares the means and medians of the two main measures of stock price crash risk. *NCSKEW* is

significantly lower in the high-risk-disclosure subsamples and *DUVOL* shows a qualitatively similar pattern. This provides preliminary evidence that higher risk disclosure is associated with lower crash risk.

Table 1 Panel C shows the yearly distribution of our final sample and the means of the main dependent and independent variables over the years. Both *NCSKEW* and *DUVOL* increase during 2007 and 2008, possibly due to the financial crisis and the economic recession in the same period. These results show that these variables are not spuriously correlated with time—that is, it is not a time-based growth effect that is causing all the main variables to move in the same direction. *RFD* slightly increases over the sample period, while *NCSKEW* and *DUVOL* are both volatile over the sample period and show no clear patterns. Regardless, to control for any time-based effects, all our regressions include year-fixed effects.

4 Research Design and Results

Our main regressions test the relationship between the risk disclosure metric, *RFD*, and crash risk metrics, *NCSKEW* and *DUVOL*. In these panel regressions, we test the following model:

$$\{Crash Risk_{it}\} = \alpha + \beta RFD_{it-1} + \gamma \{Crash Risk_{it-1}\} + \delta X_{it-1} + \varphi Year FE + \omega Ind FE + \varepsilon \quad (4)$$

The dependent variable $\{Crash Risk_{it}\}$ is one of the crash risk measures, *NCSKEW* and *DUVOL*; RFD_{it-1} is risk factor disclosure for firm i at time $t - 1$ and $\{Crash Risk_{it-1}\}$ includes the lagged crash risk value to control for past crash risk. X_{it-1} is a vector of control variables at time $t - 1$. φ and ω control for year and two-digit SIC code industry fixed effects, respectively, thereby accounting for any unobservable time or industry effects such as trends, industry shocks, and the like. Standard errors are clustered at firm level.

Following the prior literature on crash risk (Chen et al. 2001; Hutton et al. 2009, Kim et al. 2011), we control for firm characteristics, including firm size (*Lmv*), market-to-book ratio (*MB*), profitability (*ROA*), and leverage (*Lev*). We further control for the following variables also found to determine crash risk: First, the variable *Dturn* is the detrended average monthly stock turnover in a year. This is a proxy for differences of opinion among investors. Chen et al. (2001) find this detrended turnover variable to be positively related to future crash risk. Second, the variable *Sigma* is the standard deviation of firm-specific weekly returns over the fiscal year period $t - 1$. More volatile stocks are more likely to experience stock price crashes in the future. Third, the variable *Ret* is defined as the arithmetic average of firm-specific weekly returns in a year. Stocks with high past returns are more likely to crash. Fourth, the variable *Accm* is a measure of accrual manipulation used in Hutton et al. (2009) and is calculated by a three-year moving sum of absolute discretionary accruals. Hutton et al. find a positive relation between *Accm* and crash risk. Fifth, *FileSize* (log of the 10-K file size in megabytes) is included to control for the readability of the document; larger documents are more difficult for investors to understand (Loughran and McDonald 2014) and are positively related to crash risk (Ertugrul et al. 2017)⁷. Finally, the lagged crash risk measure ($NCSKEW_{t-1}$ or $DUVOL_{t-1}$) is included as a control because Chen et al. (2001) find that it is correlated with crash risk in year t .

⁷ Our results are similar when an alternative measure of readability, total words of 10-Ks (Li 2008), is used instead.

4.1 Empirical Results

4.1.1 Baseline Regressions Results

Table 2 reports the baseline results of equation (4) for the relationship between *RFD* and stock price crash risk. The dependent variable is *NCSKEW* in the odd columns and *DUVOL* in the even columns. The coefficient of *RFD* is negative and statistically significant, suggesting that firms with more *RFD* are associated with lower stock price crash risk in the future.

The results are economically significant as well. Using the coefficient of *RFD* (-0.019) in column (1), an increase in *RFD* by one standard deviation (1.029) is associated with a decrease in the negative skewness by $1.029 \times (-0.019) = -0.020$. In comparison, a one-standard-deviation decrease in *MB* is associated with a decrease in the negative skewness by $-4.366 \times 0.003 = -0.013$, which is less than the change in skewness for *RFD*.⁸ Furthermore, given that the median negative skewness in our sample is -0.027 , the effect of *RFD* on the stock price crashes is both statistically significant and economically meaningful, which is also true for *DUVOL*, as shown below.

Column (2) of Table 2 shows the regression results of *DUVOL*. The coefficient of *RFD* is again negative and statistically significant, suggesting that, *ceteris paribus*, firms with greater risk factor disclosure are associated with fewer downside deviations. Economically, the estimates in column (2) suggest that a one-standard-deviation increase in the *RFD* (1.029) is associated with a $1.029 \times (-0.005) = -0.005$ decrease in the downside deviations, which translates to a 7.25%

⁸ Chang et al. (2017) show a one-standard-deviation increase in *MB* exacerbates negative skewness by 0.018. Our magnitude is comparable.

decrease relative to the mean of *DUVOL* (-0.071). Overall, the results suggest that stock price crash risk decreases with the percentage of risk keywords in the disclosure.

The estimated coefficients of the control variables also exhibit the expected signs and are largely consistent with prior studies. For instance, we find that the average monthly share turnover ($DTURN_{t-1}$), stock price skew in the past year ($NCSKEW_{t-1}$), and the average stock return (RET_{t-1}) are positively related to crash risk as measured by either *NCSKEW* or *DUVOL*. Consistent with Chen et al. (2001) and Hutton et al. (2009), we also find that a firm's market size and growth opportunity proxied by the market-to-book ratio are positively associated with stock price crash risk and that leverage and sigma (volatility) are negatively associated with crash risk.

Some firms may voluntarily provide risk factor disclosures prior to the SEC's mandate of risk factor disclosures in 2005. We explore whether these voluntary disclosures are more informative than mandatory disclosures and whether firms provide more information after the mandate by creating *Post2005*, a dummy variable equal to one to indicate the period after 2005 (mandatory period) and zero otherwise, and we interact *Post2005* with our main variable of interest *RFD*. As can be seen in columns (3) and (4), the coefficient of *RFD* is not significant and thus has no statistically measurable impact on the two proxies for stock price crash risk, *NCSKEW* and *DUVOL*. Namely—and as hypothesis 2 suggests—in the voluntary period, RFDs have no impact on crash risk. However, $RFD \times Post2005$ has a significantly negative coefficient for both *NCSKEW* and *DUVOL*. This shows that the results are not due to voluntary disclosure; it is the mandating of risk factor disclosures that appears to reduce stock price crash risk.

There are fewer observations in the first two years (1996 and 1997) and the last year (2013) of our sample period. Therefore, in columns (5) to (8), we repeat the regressions in columns (1) to

(4) for a narrower sample period that spans from 1998 to 2012. The results remain consistent and are qualitatively similar in this alternative sample period.

[Table 2 about here]

4.1.2 Identification Strategies

It is possible that the causal effect of risk factor disclosures on future stock price crash risk runs from crash risk to RFDs (i.e., reverse causality) or that they are simultaneously determined by an omitted variable. For instance, greater ex-ante risk could prompt decreased RFDs simply because the managers engage in hiding bad news. On the other hand, RFDs could be associated with overall corporate transparency, and therefore other elements of the information environment may instead cause the decrease in crash risk. To alleviate these concerns, we use two SEC regulatory changes that took effect in 2005 and 2008 as exogenous shocks to isolate the effect. The first regulatory change by SEC is the 2005 mandate that firms include an Item 1A risk factors section in their 10-K filings. The second regulatory change exempted eligible firms from doing so.

A. Differences-in-Differences: US vs. Canadian Firms around RFD Mandate

We first use the 2005 SEC mandate of risk disclosures in 10-K filings as a quasi-natural experiment. Risk disclosure was optional prior to 2005 but became mandatory for US-listed firms after 2005, but there was no change in regulation for Canadian firms. Following Chiu et al. (2018), we use Canadian firms that are not affected by the regulatory change in RFDs in 2005 as a control sample. We first use propensity score matching to construct a matched sample of Canadian and US firms based on weekly returns (*RET*), firm size (*Lmv*), profitability (*ROA*), and industry.⁹ We

⁹ We conduct the matching with replacement, using 0.25 times the standard deviation of the estimated propensity scores (approximately 1.5%) as the caliper.

also exclude from the Canadian sample any firms that are cross-listed in the USA and Canada since those firms would be subject to SEC rules. We then use this matched sample to set up a DiD analysis. Table 3 Panel A shows that US-listed firms (treatment group) experienced a significant decline in crash risk following the 2005 SEC mandate relative to Canada-listed firms (control group). The results provide additional evidence that mandating risk disclosure mitigates stock price crash risk.

To ensure the robustness of our DiD results, we perform a placebo (falsification) test and check if the parallel-trends assumption holds. First, to verify that our results are indeed driven by RFDs and not more general time-series trends unrelated to information disclosures, we assume a pseudoshock in a random year other than 2005, such as 2002, and conduct a placebo test following the same DiD analysis procedure as above. We expect to see no effect in this placebo test. Table 3 Panel B reports the results. The coefficients of the interaction term, $Treat \times Post2002$, in columns (1) and (2) are both statistically insignificant, confirming our prediction. There is no difference in crash risk between US- and Canada-listed firms in the 2002 period because there was no mandate for additional risk factor disclosures.

Second, we test parallel trends upon which the DiD methodology is based. That is, we test whether unobserved characteristics vary over time between the treatment and control groups. To test for pre-event parallel trends in the outcome variables, we run regressions of the dependent variables on the treatment variable, time dummies for all pre-event periods, their interactions, and all the control variables used in our baseline regressions, using pre-event data only. If the trends are not parallel, at least one of the coefficients on the interaction terms would be statistically significant in these tests. Table 3 Panel B (columns 3 and 4) reports the results. All estimates on the interaction terms are not significantly different from zero for both the *NCSKEW* and *DUVOL*

specifications. While the identification assumption is fundamentally untestable since we do not observe the counterfactual of the 2005 SEC mandate not having been issued, our test results suggest that the parallel trends assumption is likely to hold.¹⁰

[Table 3 about here]

To visualize the parallel trends and treatment effects, we graph the crash risk before and after the 2005 mandate for both the treatment and control groups. Figure 1 shows that before 2005, there are fairly parallel trends between the two groups; after 2005, the crash risk for the control group rises while the treated sample—which had to disclose its risk factors—saw relatively flat levels of *NCSKEW* and *DUVOL*, suggesting the existence of treatment effects. Overall, these findings are consistent with our conjecture that the regulatory mandate strengthens the effectiveness of RFDs on mitigating stock price crash risk.

[Figure 1 about here]

B. Regression Discontinuity Design (RDD): Smaller Reporting Company Exemption

In 2008, the SEC expanded the definition of smaller reporting companies and exempted a wide swath of firms from the Item 1A risk disclosure section requirement.¹¹ Specifically, the 2008 SEC exemption is allowed for smaller reporting companies with a higher threshold (below \$75

¹⁰ Though the Sarbanes Oxley (SOX) regulation took place in a similar time frame to the Item 1A mandate, there are significant differences in the timelines. For example, SOX Section 404 came into effect for fiscal year 2004, whereas the Item 1A requirement came into effect for fiscal year 2005, leaving a one-year gap between the two. Furthermore, the parallel trends test in Table 3 Panel B shows there is no significant difference in the *Year2004xTreat* and *Year2005xTreat* coefficients—i.e., the SOX implementation in 2004 did not affect the parallel trends for stock price crash risk.

¹¹ Prior to 2008, smaller reporting companies were only limited to firms with a public equity float <\$25 million and revenue <\$25 million. See <https://www.federalregister.gov/documents/2008/01/04/E7-24965/smaller-reporting-company-regulatory-relief-and-simplification>.

million in public equity float). At the same time, the arbitrary nature of the SEC cutoff suggests that being just above or below the cutoff is locally random; that is, firms around the cutoff are similar across all characteristics, except, potentially, for the disclosure of Item 1A. Therefore, any discontinuous changes in the stock price crash risk below the cutoff could be attributed to the discontinuity in the mandate for disclosure of RFDs. This cutoff allows us to use a regression discontinuity design (RDD) to exploit this natural experiment (Chava and Roberts 2008; Iliev 2010; Gustafson and Iliev 2017), where very similar firms are randomly sorted into groups of treated (no mandatory Item 1A disclosure) and untreated groups (with mandatory Item 1A disclosure).¹²

Though firms could optionally have risk disclosures, the optional risk disclosures would not have been subject to the same SEC oversight and rules, meaning that there would be no guarantee that the risk disclosures would include any material or timely information.

The regressions setup for this sharp RDD are as follows:

$$\{Crash Risk_{it}\} = \alpha + \beta_1 SB_{it} + \delta X_{it-1} + \varphi Year FE + \omega Ind FE + \epsilon \quad (5)$$

Our main specification is a local linear regression estimation on narrow bandwidth around the \$75 million public equity float cutoff. We include only firms that are near the discontinuity in the regression and ensure they are similar across various characteristics except for the mandate of risk factor disclosure. Further, as the SEC regulation change becomes effective in 2008, we include

¹² There was an exemption for SOX Section 404 for smaller reporting companies; however, the timing between this and the RFD exemption is slightly different. Smaller reporting companies were exempt from SOX Section 404's Management Report (MR) until fiscal year 2007 and from the rest of Section 404 until fiscal year 2010. In contrast, the RFD exemption began in 2008, a year after the MR and 2 years before the rest of Section 404 exemptions. Consequently, the RDD does not have any conflating events.

observations only after 2008. *SB* is an indicator variable equal to one if the firm whose public equity float is below \$75 million.¹³

Figure 2 plots firms whose public equity float is within \$5 million of the \$75 million cutoff (\$70 and \$80 million) during the post-2008 period. The figure shows local sample means (i.e., the dots in the graph) of the RFD using nonoverlapping evenly spaced bins on each side of the \$75 million threshold. The dashed vertical line in the plot represents the \$75 million threshold, and the solid lines are best-fit regressions estimated separately within \$5 million on each side of the threshold. The plot shows a discontinuous jump in RFDs around the \$75 million threshold.

[Figure 2 about here]

We report the regression results in Table 4. The sample includes observations only in years 2008 and after (when the exemption is allowed). The dependent variable is *NCSKEW* in the odd columns and *DUVOL* in the even columns. *SB* is an indicator variable equal to one if the firm's public equity float is below \$75 million. The discontinuity sample in Panels A, B, and C includes firms whose public equity float is within \$5 million (\$70 and \$80 million), \$7.5 million (\$67.5 and \$82.5 million), and \$10 million (\$65 and \$85 million) of the \$75 million cutoff, respectively.

Considering the smaller sample size in this analysis, we start with a number of model specifications with limited controls to increase the degree of freedom for a valid test. For instance, we include only the *SB* dummy variable in columns (1) and (2), add *Lmv* and lag crash risk in columns (3) and (4). Columns (5) and (6) then report the regression results of model (5).

[Table 4 about here]

¹³ Smaller reporting companies are determined the same year as the firm is filing. Thus, a firm with market cap of \$74 million at the end of fiscal year 2011 would be eligible to be a smaller reporting company for FY2011.

In all the columns, *SB* has a significantly positive coefficient. Hence, the results suggest that smaller reporting companies experience greater negative skewness and larger downside deviations because they are exempted from including a risk disclosure section in their 10-K filings. Conversely, when a firm is required to have a risk disclosure section (because it no longer meets the definition of a smaller reporting company), its chance of stock price crash risk is reduced. Taken together with the results from the previous section, these results provide further evidence that the mandatory inclusion (exemption) of RFDs reduces (increases) stock price crash risk. These results could also be viewed as partial evidence that the effect of RFDs on the reduction in the stock price crash is not plagued by endogeneity problems.

4.2 Moderators

In this section, we test Hypothesis 3a-c and examine possible moderators which impact the strength of the relationship between RFDs and stock price crash risk.

4.2.1 Information Asymmetry

We first investigate whether information asymmetry is related to the impact of RFDs on stock price crashes. In the spirit of Jin and Myers (2006), stock price crash risk is related to information asymmetry because managers are able to hoard bad news and not release it until they can hoard it no longer. Enhanced disclosure should combat these problems and improve the information environment surrounding the firm. Logically, we would expect risk disclosure to have a greater effect mitigating subsequent crash risks for firms with high information asymmetry.

We test Hypothesis 3a with three proxies for information asymmetry at the firm level. First, following Zhang (2006) and Chiu et al. (2018), we use analysts' forecast earnings dispersion to capture the information uncertainty resulting from the volatility of the firm's fundamentals. We

collect the analyst earning forecast data from I/B/E/S summary dataset and construct a dispersion of analyst earnings forecasts (*ANADISP*) before the last year end measured by the standard deviation of these forecasts. Second, to capture the complexity of a firm's business operations, we use the number of business segments (*SEG*) that a firm has in different industries. When the firm's underlying fundamentals and business operations are complex, it is more difficult for market participants to assess the value or prospects of the firm precisely. The firm is subject to greater information uncertainty. The business segment information is obtained from the Compustat Industrial Segment (CIS) database. Last, motivated by a vast literature in market microstructure, we use the average daily posted bid and ask spread (*BAS*) as a trade-based measure to proxy information asymmetry. We revise equation (4) and add an interaction term between *RFD* and each of the information asymmetry proxies to run regressions.

We present the results in Table 5. Results of *NCSKEW* are reported in the odd columns and those of *DUVOL* in the even columns. For brevity, we only report the coefficients of key variables. In five out of six model specifications, the coefficients on the interaction term are negative and significant, suggesting that a possible channel for the *RFD* and crash risk relationship is information asymmetry. Taken together, these findings support the first part of our third hypothesis because risk factor disclosure has a larger reduction in the risk of stock price crashes when information asymmetry is higher—such as when the firm's underlying fundamentals are more volatile, when the firm's business operations are more complex, or when the stock market is more uncertain about the firms. The results indicate that the impact of risk factor disclosure on the stock price crashes is stronger for firms with greater information asymmetry.

[Table 5 about here]

4.2.2 Shareholder Power

The other potential channel for the RFD impact on stock price crash risk is shareholder power. If shareholders have more power over managers through good governance or through litigation, they are more likely to release relevant and material information and then that information will be impounded into the market, reducing the chance of a large build-up of negative news. Such inferences suggest that firms with greater shareholder power have stronger association between RFDs and subsequent crash risks than those with weaker shareholder power: RFDs released by the former should be more impactful and relevant to market participants.

To test this hypothesis, we use three proxies for shareholder power: shareholder litigation risk, entrenchment index (*E-Index*), and institutional ownership. Shareholder litigation is one way for shareholders to enforce their rights. For example, managers exposed to higher litigation risk are motivated to reveal bad news in a timelier manner (Baginski et al. 2002). If managers fail to disclose material bad news in a timely manner, they could face legal liability or owe damages to market participants (Skinner 1994). Thus, managers who face higher litigation risks would have stronger incentive to disclose relevant and material information as soon as possible to reduce the chance of lawsuits. Consequently, we expect a stronger relation between *RFD* and stock price crash risk for firms with higher litigation risk. We employ the litigation risk estimate equation proposed by Kim and Skinner (2012) and create a variable, *LitRisk*, at the firm level to interact with our key variable of interest, *RFD*.

The results are presented in Table 6 columns (1) and (2). In both columns, the coefficients on the interaction term between risk factor disclosure and litigation risk are negative and statistically significant, suggesting that mandatory risk disclosure has stronger impact on the subsequent negative skewness and downside deviations for firms with higher litigation risk. In

short, litigation risk encourages firm managers to disclose bad news earlier in mandatory RFDs to prevent costly shareholder lawsuits later. This reduction in bad news hoarding thereby reduces the risk of future stock price crashes. Thus, RFDs and litigation risk complement each other in reducing stock price crash risk.

In a similar vein, corporate governance can moderate how mandatory RFDs impact firms' stock price crash risk. As noted earlier, well-governed firms are associated with the more timely and relevant release of information. Consequently, mandating RFDs will lead these well-governed firms to disclose more useful information than poorly governed firms. This in turn will strengthen the negative relationship between RFDs and stock price crash risk for well-governed firms.

To measure corporate governance, we use both internal and external metrics. Specifically, we use *E-index*, the entrenchment index (Bebchuk et al. 2009), as the proxy for internal corporate governance. *E-index* measures the number of rules the firm has implemented to entrench management. For ease of interpretation, the *E-index* is multiplied by -1 , so lower scores represent worse governance¹⁴. The *E-index* sample ends in 2010 as *E-index* data is only available until 2009 ($E-index_{t-1}$ data is used in the regressions). We use the percentage of institutional investor ownership (*IO*) as a proxy for external governance (Chung and Zhang 2011). Institutional investors are associated with better corporate governance, so firms with high *IO* should see negative information impounded sooner.

Columns (3) and (4) show the results of regressing stock price crash risk against the interaction term between *RFD* and the *E-index*. The coefficient for the interaction is significantly negative for both *NCSKEW* and *DUVOL*. Furthermore, institutional ownership also shows a

¹⁴ (in the original measure, higher scores represent worse governance)

similar pattern to the *E-index* results. Columns (5) and (6) also show that better external governance strengthens the association between *RFD* and crash risk, while worse external governance attenuates it. Our results suggest corporate governance is a potential moderator because RFDs seem to provide market participants with more relevant and material information in well-governed firms. Taken together, risk factor disclosure becomes more useful as corporate governance improves, thus strengthening the association between *RFD* and future crashes.

[Table 6 about here]

In addition, due to the concern that our proxies imperfectly measure shareholder power, we exploit two relevant exogenous shocks. First, we use the passage of universal demand laws as an exogenous shock to the litigation risk firms face. Universal demand (UD) laws make it more difficult for shareholders to successfully litigate against firms, thereby lowering litigation risk (Nguyen et al. 2018). Because UD laws were state laws passed over time, they create a series of natural experiments over time.

Table 7 Panel A includes an interaction for UD law passage, where UD is equal to one if the firm's state has passed a UD law that year or earlier, and zero otherwise. As can be seen, the passage of a UD law is associated with a weaker effect between risk factor disclosures and stock price crash risk. Column (1) shows a directional decrease in the relationship for *NCSKEW*; column (2) shows that the relationship is significantly reduced for *DUVOL*. Thus, the passage of a UD law likely reduces the ease of shareholder litigation, which in turn reduces shareholder power. This reduction in shareholder power allows firm managers to put less useful information in the RFDs, weakening the potential effectiveness of litigation, which in turn weakens the effect of RFDs on reducing stock price crash risk.

Second, we use the adoption of business combination (BC) laws as an exogenous shock to management entrenchment. The existing evidence is largely consistent with the notion that BC laws entrench firms and increase the potential for managerial slack because takeover protection effectively shields incumbent managers from the external disciplining force of the market for corporate control and further exacerbates agency conflicts (e.g., among others, Bertrand and Mullaniathan 2003; Giroud and Mueller 2010). Consequently, entrenched managers are less likely to disclose private information, particularly negative information, weakening the effect of the RFDs on the subsequent stock crash.

We create a dummy variable *BC* to indicate the passage of BC laws as an exogenous entrenchment shock that takes the value of one in the years after firm *i*'s state of incorporation has passed BC laws, and zero otherwise.¹⁵ We then interact *BC* with our main variable of interest *RFD* and examine how governance affects the association between *RFD* and stock crash. The results are reported in Table 7 Panel B. The coefficient of the interaction term is significant and positive on the crash risk measures, suggesting that the mitigating effect of RFDs is weakened for firms incorporated in states that passed BC laws. In other words, firms with weaker governance and more entrenched managers appear to disclose less credible and less-informative risk factors, resulting in higher future stock price crashes.¹⁶

[Table 7 about here]

¹⁵ We obtain the year when the business combination law in each state was passed from Table 2 of Cain, McKeon, Solomon (2017).

¹⁶ We also run all the cross-sectional tests for the voluntary period (i.e., the pre-2005 period) and find that in the voluntary period, the impact of risk disclosures on crash risk does not depend on information asymmetry or shareholder power. The coefficients on the interaction terms are insignificant, with the tests on the average daily bid-ask spread and business combination laws as the only exceptions.

4.3 Robustness Tests

4.3.1 Alternative Measure for Stock Price Crash Risk: *CRASH_DUMMY*

In this section, we use an alternative measure of crash risk as a robustness check. Following the existing literature (e.g., Hutton et al. 2009; Chang et al. 2017; Ertugrul et al. 2017), the alternative proxy for stock price crashes is a dummy variable (*CRASH_DUMMY*) that takes the value of one if a firm experiences one or more crash weeks in a fiscal year, and zero otherwise. We define a firm's crash weeks as those during which the firm-specific return is below 3.2 standard deviations of the average firm-specific weekly returns over the entire fiscal year.¹⁷ We then reassess the baseline model of equation (4). Instead of controlling for the lagged crash dummy, we follow Chang et al. (2017) to control for the lagged negative skewness (*NCSKEW*).

To simplify the interpretation of the estimates, we check the marginal effects of each explanatory variable at the mean value from the logit regression model and the results of a linear probability model (LPM). The coefficients of *RFD* in both model specifications are negative and statistically significant, suggesting risk disclosure is negatively associated with subsequent crash risk. Overall, our results are robust to the various measures of stock price crashes and suggest that the SEC mandate of risk factor disclosure improves the information environment and mitigates subsequent crashes.

4.3.2 Other Robustness Tests

In one robustness test, we regress the risk disclosure against the firm characteristics identified by Campbell et al. (2014) and then use the residuals from the first regression to predict the impact

¹⁷ In an unreported result, we find that, on average, 18.1% of firm-years in our sample experience one or more crash weeks during the fiscal year. This figure is generally in line with those reported in existing research (e.g., Hutton et al. 2009; Chang et al. 2017).

of these orthogonalized values on stock price crash risk for the firm in equation (4). Our results (untabulated) are similar, suggesting that our results are not due to firm characteristics.

We also test the concern that disclosures, including RFDs, may be boilerplate and sticky (nearly identical from year to year) and thus not informative. We rerun Table 2 with the changes in *RFD* instead of the absolute level of *RFD*. By using changes in *RFD* we can neutralize any issues with boilerplate disclosures, which should be constant and therefore netted out by our changes. In untabulated tests, we find that the results of Table 2 are qualitatively similar—changes in RFDs still negatively predict stock price crash risk. This shows that the results are not affected by boilerplate disclosures.

There is also a concern that our results may be affected by the dot-com crash or the global financial crisis. To address this, we run regressions where we include a control variable for the dot-com crash and the financial crisis. In untabulated tests, for all three metrics of crash risk, *NCSKEW*, *DUVOL*, and *CRASH_DUMMY*, the impact of *RFD* remains significant after controlling for the crisis period and an interaction between it and *RFD*. Consequently, the effect of risk disclosures on crash risk is unlikely to be due to these periods of market distress.

5 Conclusion

This paper demonstrates that mandated, but not voluntary, risk factor disclosure is associated with reduced stock price crash risk. Furthermore, we find that the identified effect is linked to information asymmetry and shareholder power. Using various proxies to capture information asymmetry, we show that firms with higher information asymmetry see a larger stock price crash risk reduction per unit of risk factor disclosure. In addition, shareholder power moderates the impact of RFDs on crash risk. Firms more vulnerable to shareholder litigation also see an increased

effect of risk factor disclosure on reducing crash risk—suggesting firms with high litigation risk are strongly motivated to use the risk factor disclosure to ensure negative information is released early. Finally, firms with better governance have RFDs that are more likely to reduce stock price crash risk than firms with poor governance.

Our findings have several implications for academics and practitioners. For academics, this research further shows that RFDs improve firms' information environment. It also lends credence to the importance of disclosures in managing the negative skewness of stock returns. For executives, this work shows the importance of an effective risk factor disclosure—enhanced disclosure early can reduce large negative events in the future. Moreover, this research lends support to the SEC and other regulatory agencies' drive to increase disclosure. Enhanced disclosure may improve firms' information environment, leading to better outcomes for firms and market participants alike.

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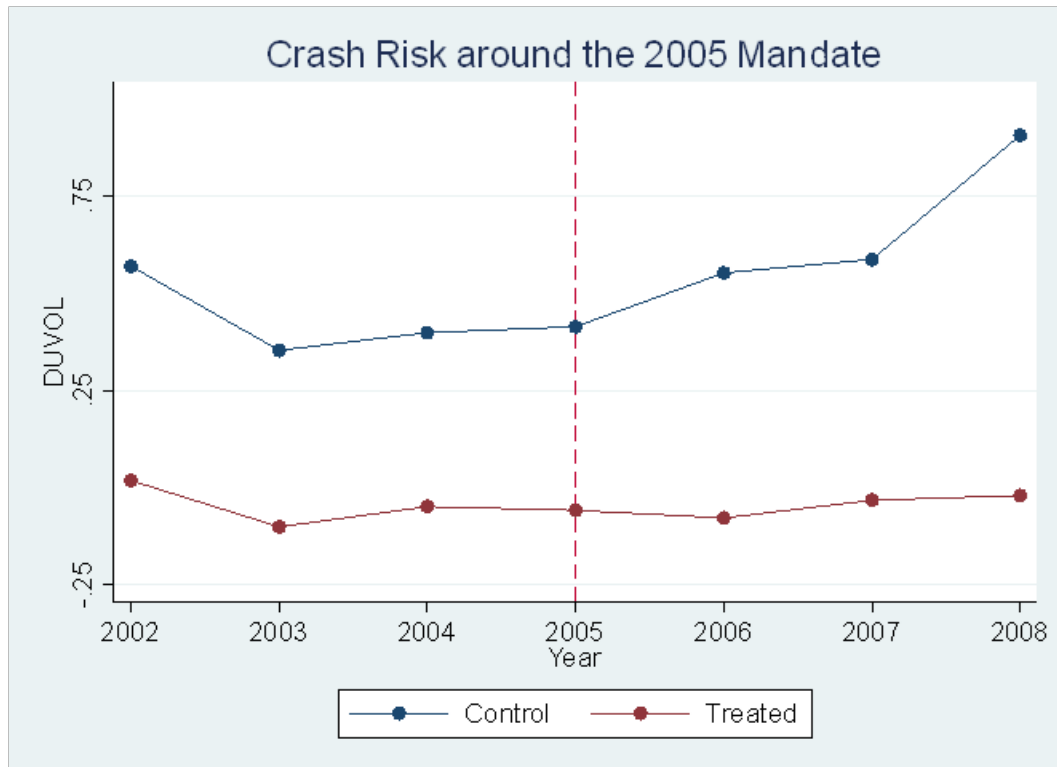
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Figure 1. Crash Risk Around the 2005 Mandate

This figure shows the level of *DUVOL* and *NCSKEW* before and after 2005 in panels A and B, respectively. *Treat* is an indicator variable equal to one for firms listed in the US (i.e., the treatment group) and zero for firms listed in Canada and not listed in the US (i.e., the control group). Firms that are cross-listed in the US are excluded from the Canadian sample. Canadian firms are matched to the US sample based on weekly returns (*RET*), firm size (*Lmv*), profitability (*ROA*), and industry.

Panel A: DUVOL



Panel B: NCSKEW

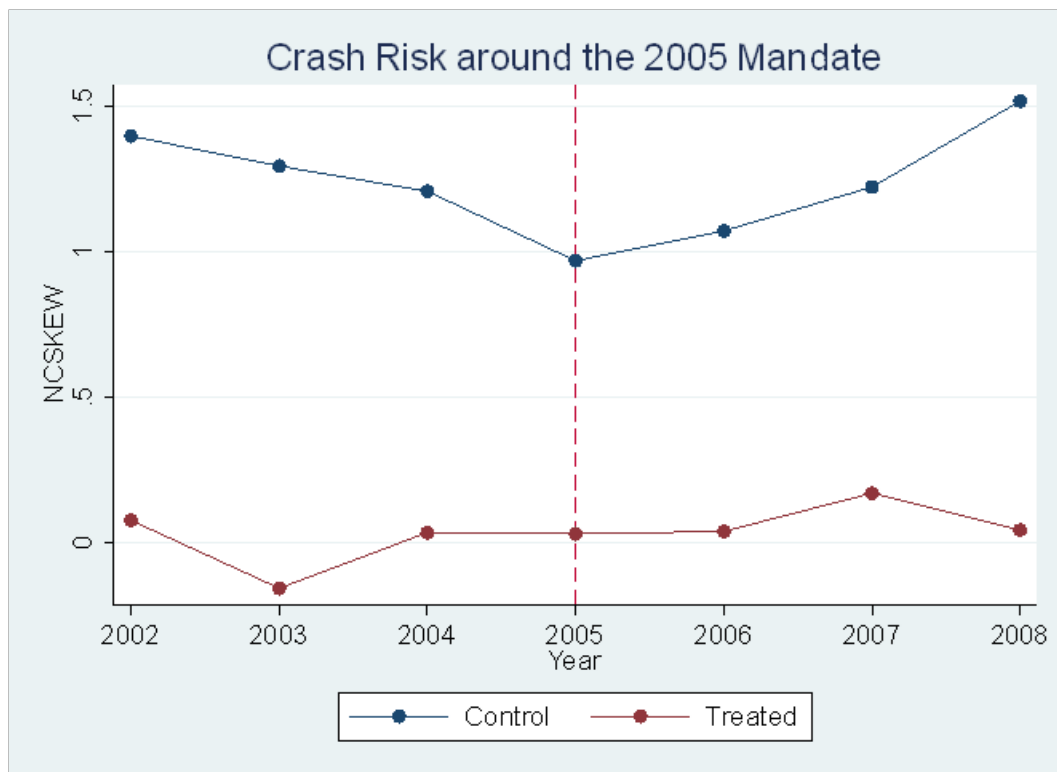


Figure 2: Risk Factor Disclosure around the \$75 million threshold

The sample consists of observations between \$70 million and \$80 million of public equity float during the post-2008 period when the SEC instituted the smaller reporting company exemption. The figure presents a plot of local sample means (i.e. the dots in the graph) of the RFD using non-overlapping evenly spaced bins on each side of the \$75 million threshold. The dashed vertical line in the plot represents the \$75 million threshold, and the solid lines are the best-fit regressions estimated separately within \$5 million on each side of the threshold.

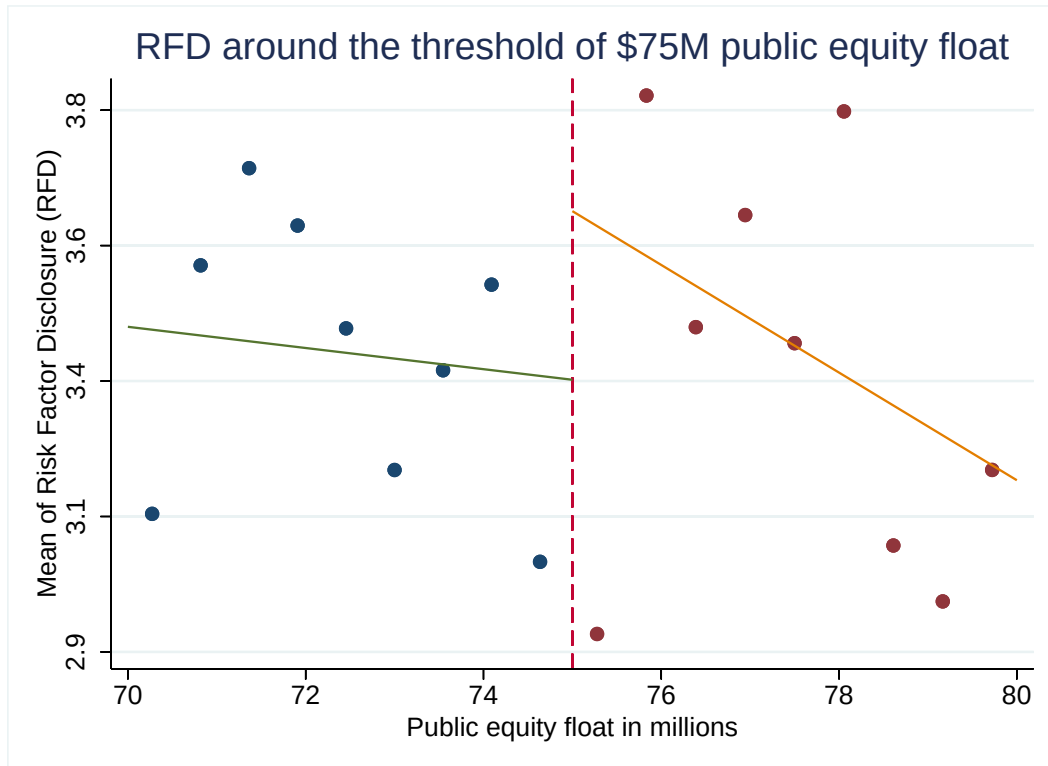


Table 1. Descriptive Statistics

This table reports the summary statistics for the final sample with 5,784 unique firms during the period 1996 to 2013. Panel A reports summary statistics for the firms' characteristics. Panel B shows the mean and median value of stock price crashes measures (*NCSKEW* and *DUVOL*) for the subsample of risk disclosure (*RFD*) above and below the median value each year. Panel C presents the yearly distribution of observations and the descriptive statistics for *RFD* and the annual stock price crashes. Panel D presents the correlations of the most relevant variables; the values in bold are significant at the 5% level. All continuous variables are winsorized at the 1st and 99th percentiles each year. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively, for *t*-tests for means and Wilcoxon tests for medians. Variables are defined in Appendix A.1.

Panel A: Total Sample Statistics

Variable	MEAN	STD	P25	MEDIAN	P75
<i>RFD</i>	3.278	1.029	2.639	3.367	3.970
<i>NCSKEW</i>	-0.003	1.120	-0.697	-0.027	0.637
<i>DUVOL</i>	-0.071	0.405	-0.335	-0.075	0.179
<i>Mv</i>	3471	16055	91	388	1558
<i>Lmv</i>	5.986	2.011	4.511	5.960	7.351
<i>MB</i>	2.933	4.366	1.212	1.999	3.440
<i>ROA</i>	0.088	0.177	0.059	0.116	0.172
<i>Lev</i>	0.206	0.202	0.011	0.170	0.330
<i>Ret</i>	-0.002	0.010	-0.007	-0.002	0.003
<i>Sigma</i>	0.065	0.034	0.040	0.057	0.082
<i>Accm</i>	0.205	0.177	0.094	0.157	0.254
<i>Dturn</i>	0.002	0.095	-0.027	0.000	0.028
<i>Filesize</i>	0.385	0.314	0.140	0.310	0.560

Panel B: Summary Statistics by RFD_{t-1}

	Below Mean RFD_{t-1}	Above Mean RFD_{t-1}	t-stat (above- below mean)	Below Median RFD_{t-1}	Above Median RFD_{t-1}	t-stat (above-below median)
<i>NCSKEW</i>	0.012	-0.019	2.79***	-0.021	-0.032	2.20**
<i>DUVOL</i>	-0.068	-0.075	1.59	-0.074	-0.076	1.03

Panel C: Mean Sample Statistics by Year

Year	N	<i>RFD</i>	<i>NCSKEW</i>	<i>DUVOL</i>
1996	366	1.084	-0.023	-0.138
1997	995	2.188	0.037	-0.140
1998	1,766	2.519	0.061	-0.141
1999	2,011	2.954	-0.230	-0.192
2000	1,921	2.866	-0.075	-0.057
2001	1,934	2.900	-0.118	-0.053
2002	2,052	3.143	0.062	0.004
2003	2,198	3.309	-0.209	-0.117
2004	2,338	3.468	0.023	-0.052
2005	2,339	3.529	0.021	-0.059
2006	2,396	3.263	0.030	-0.080
2007	2,355	3.306	0.162	-0.040
2008	2,293	3.470	0.056	-0.019
2009	2,469	3.660	-0.215	-0.131
2010	2,494	3.650	-0.075	-0.100
2011	2,412	3.639	0.080	-0.044
2012	2,352	3.665	0.195	0.003
2013	276	3.488	0.150	-0.013

Table 2. Risk Disclosure and Crash Risk Multivariate Regressions

This table presents the results of the impact of risk disclosure on stock price crash risk. The dependent variable is *NCSKEW* in the odd columns and *DUVOL* in the even columns. In columns (1) to (4), the results include observations in the sample during the period 1996 to 2013; in columns (5) to (8), we report results during the period 1998 to 2012. *Post2005* is a dummy variable equal to one for the period after 2005 and zero otherwise; all continuous variables are winsorized at the 1st and 99th percentiles. The constant term is not reported for brevity. *t*-statistics based on robust standard errors clustered by firm are reported in parentheses below the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. Variables are defined in Appendix A.1.

Variables	1996-2013				1998-2012			
	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>RFD_{t-1}</i>	-0.019*** (-2.607)	-0.005* (-1.932)	0.001 (0.073)	0.000 (0.059)	-0.020*** (-2.757)	-0.005* (-1.825)	0.000 (0.007)	0.001 (0.357)
<i>RFD_{t-1} × Post2005</i>			-0.064*** (-4.962)	-0.018*** (-3.821)			-0.065*** (-4.966)	-0.020*** (-4.143)
<i>Post2005</i>			-0.008 (-0.018)	0.110 (0.932)			0.398*** (6.624)	0.131*** (6.037)
<i>NCSKEW_{t-1}</i>	0.060*** (8.240)		0.060*** (8.213)		0.060*** (8.102)		0.060*** (8.076)	
<i>DUVOL_{t-1}</i>		0.055*** (9.097)		0.055*** (9.086)		0.055*** (8.813)		0.054*** (8.801)
<i>Lmv_{t-1}</i>	0.069*** (16.062)	0.025*** (15.088)	0.071*** (16.283)	0.025*** (15.239)	0.070*** (15.781)	0.025*** (14.768)	0.072*** (15.986)	0.025*** (14.922)
<i>MB_{t-1}</i>	0.003** (1.964)	-0.000 (-0.146)	0.003* (1.767)	-0.000 (-0.301)	0.003* (1.931)	-0.000 (-0.197)	0.002* (1.734)	-0.000 (-0.365)
<i>ROA_{t-1}</i>	0.155*** (3.615)	0.128*** (8.471)	0.160*** (3.751)	0.129*** (8.601)	0.150*** (3.433)	0.125*** (8.173)	0.156*** (3.569)	0.127*** (8.315)
<i>Lev_{t-1}</i>	-0.168*** (-4.757)	-0.050*** (-3.851)	-0.156*** (-4.440)	-0.047*** (-3.608)	-0.169*** (-4.706)	-0.050*** (-3.769)	-0.158*** (-4.393)	-0.046*** (-3.512)
<i>Ret_{t-1}</i>	13.630*** (18.612)	4.375*** (19.448)	13.630*** (18.624)	4.378*** (19.467)	13.705*** (18.399)	4.338*** (18.923)	13.708*** (18.420)	4.343*** (18.949)
<i>Sigma_{t-1}</i>	-0.776*** (-3.158)	-0.437*** (-4.926)	-0.798*** (-3.250)	-0.444*** (-5.006)	-0.826*** (-3.312)	-0.468*** (-5.191)	-0.852*** (-3.418)	-0.477*** (-5.290)
<i>Accm_{t-1}</i>	0.048 (1.327)	0.009 (0.702)	0.042 (1.173)	0.007 (0.579)	0.044 (1.198)	0.007 (0.530)	0.038 (1.044)	0.005 (0.395)
<i>Dturn_{t-1}</i>	0.185*** (3.072)	0.050** (2.297)	0.186*** (3.097)	0.050** (2.317)	0.162*** (2.662)	0.040* (1.810)	0.163*** (2.682)	0.040* (1.828)
<i>LnFileSize_{t-1}</i>	0.019 (1.163)	-0.004 (-0.527)	0.024 (1.477)	-0.003 (-0.385)	0.018 (1.081)	-0.003 (-0.382)	0.023 (1.385)	-0.002 (-0.234)
Constant	0.064 (0.138)	-0.338** (-2.405)	0.047 (0.100)	-0.352** (-2.473)	-0.108** (-2.095)	-0.252*** (-3.138)	-0.349*** (-6.116)	-0.345*** (-4.248)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	34,951	34,951	34,951	34,951	33,312	33,312	33,312	33,312
Adjusted R ²	0.048	0.064	0.049	0.064	0.049	0.064	0.050	0.064

Table 3. Difference-in-Differences (DiD): US (Treatment) vs. Canadian (Control) Firms

Panel A of this table presents the results of the DiD analysis. *Post2005* is a dummy variable equal to one for the period after 2005 and zero otherwise. *Treat* is an indicator variable equal to one for firms listed in the US (i.e., the treatment group) and zero for firms listed in Canada and not in the US (i.e., the control group). Firms that are cross-listed in the US are excluded from the Canadian sample. Canadian firms are matched to the US sample based on weekly returns (*RET*), firm size (*Lmv*), profitability (*ROA*), and industry. Panel B presents the results of placebo tests and a parallel trends test. *Treat* is an indicator variable equal to one for US firms (i.e., the treatment group) and zero for non-cross-listed Canadian firms (i.e., the control group). In Panel A, *Post2002* is a dummy variable equal to one for the period after 2002 and zero otherwise. In Panel B, *YearN* is a dummy variable equal to one for the year equal to *N* and zero otherwise. The constant term is not reported for brevity. Controls are the same as those in Table 2. *t*-statistics based on robust standard errors clustered by firm are reported in parentheses below the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. Variables are defined in Appendix A.1.

Panel A. DiD Results

	<i>NCSKEW</i>	<i>DUVOL</i>
	(1)	(2)
<i>Post2005</i> × <i>Treat</i>	−0.120*** (−4.99)	−0.028** (−2.29)
<i>Treat</i>	0.916*** (43.37)	−0.146*** (−15.40)
Controls	Yes	Yes
Year FE	Yes	Yes
Industry FE	Yes	Yes
Observations	44,205	44,224
Adjusted R ²	0.127	0.218

Panel B. Placebo and Parallel Trends Tests

	Placebo tests		Parallel trends tests	
	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>
	(1)	(2)	(3)	(4)
<i>Post2002</i> × <i>Treat</i>	0.180 (1018)	−0.007 (−0.09)		
<i>Treat</i>	0.816*** (5.60)	−0.147** (−2.02)	3.771 (1.47)	−0.328 (−0.94)
<i>Year1998</i> × <i>Treat</i>			0.519 (0.75)	−0.303 (−0.85)
<i>Year1999</i> × <i>Treat</i>			0.367 (0.55)	−0.309 (−0.87)
<i>Year2000</i> × <i>Treat</i>			0.435 (0.59)	−0.241 (−0.63)
<i>Year2001</i> × <i>Treat</i>			−0.338 (−0.40)	0.064 (0.15)
<i>Year2002</i> × <i>Treat</i>			0.635 (0.94)	−0.396 (−1.11)
<i>Year2003</i> × <i>Treat</i>			0.284 (0.38)	−0.143 (−0.36)
<i>Year2004</i> × <i>Treat</i>			−0.152 (−0.15)	0.073 (0.15)
<i>Year2005</i> × <i>Treat</i>			0.444 (0.66)	−0.118 (−0.33)
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes
<i>Observations</i>	44,274	44,256	24,631	24,629
<i>Adjusted R²</i>	0.218	0.134	0.191	0.144

Table 4. Risk Disclosure and Crash Risk: Regression Discontinuity Design

This table reports the results of Regression Discontinuity Design (RDD) when the exemption for risk factor disclosures was extended to smaller reporting companies in 2008. The sample includes observations only in years 2008 and after (when the exemption is allowed). The dependent variable is *NCSKEW* in the odd columns and *DUVOL* in the even columns. *SB* is an indicator variable equal to one if the firm whose public equity float is below \$75 million. The discontinuity sample in Panel A, B, and C includes firms whose public equity float is within \$5 million (\$70 and \$80 million), \$7.5 million (\$67.5 and \$82.5 million), and \$10 million (\$65 and \$85 million) of the \$75 million cutoff, respectively. The constant is not reported for brevity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. Variables are defined in Appendix A.1.

<i>Panel A: Bandwidth within +/- \$5M of \$75M public floating</i>						
	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>
	(1)	(2)	(3)	(4)	(5)	(6)
SM_{t-1}	0.441*** (2.720)	0.152** (2.523)	0.456** (2.375)	0.165** (2.246)	0.353* (1.751)	0.127* (1.669)
$NCSKEW_{t-1}$			-0.101 (-1.195)		0.070 (0.698)	
$DUVOL_{t-1}$				-0.055 (-0.627)		0.079 (0.863)
Lmv_{t-1}			0.160 (0.963)	0.045 (0.707)	0.304 (1.358)	0.115 (1.379)
Constant	-0.448*** (-3.753)	-0.241*** (-5.373)	-1.191 (-1.416)	-0.460 (-1.425)	-1.465 (-1.194)	-0.642 (-1.411)
Controls	No	No	No	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes	Yes
Ind FE	No	No	No	No	Yes	Yes
Observations	211	211	210	210	188	188
Adjusted R2	0.030	0.025	0.023	0.011	0.033	0.036
<i>Panel B: Bandwidth within +/- \$7.5 M of \$75M public floating</i>						
	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>
	(1)	(2)	(3)	(4)	(5)	(6)
SM_{t-1}	0.541*** (3.949)	0.167*** (3.397)	0.472*** (2.963)	0.153*** (2.610)	0.358** (2.272)	0.117** (2.062)
$NCSKEW_{t-1}$			-0.063 (-0.933)		0.095 (1.129)	
$DUVOL_{t-1}$				-0.025 (-0.373)		0.086 (1.217)
Lmv_{t-1}			0.115 (0.848)	0.043 (0.875)	0.158 (1.060)	0.078 (1.463)
Constant	-0.476*** (-4.625)	-0.242*** (-6.549)	-0.971 (-1.423)	-0.438* (-1.731)	-0.819 (-1.009)	-0.487* (-1.688)
Controls	No	No	No	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes	Yes
Ind FE	No	No	No	No	Yes	Yes
Observations	316	316	315	315	288	288
Adjusted R2	0.045	0.033	0.059	0.033	0.059	0.056

Panel C: Bandwidth within +/- \$10M of \$75M public floating

	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>
	(1)	(2)	(3)	(4)	(5)	(6)
SM_{t-1}	0.510*** (3.825)	0.161*** (3.319)	0.437*** (2.771)	0.152** (2.586)	0.320** (2.050)	0.116** (2.009)
$NCSKEW_{t-1}$			-0.053 (-0.823)		0.109 (1.354)	
$DUVOL_{t-1}$				-0.015 (-0.232)		0.099 (1.418)
Lmv_{t-1}			0.091 (0.708)	0.039 (0.820)	0.130 (0.904)	0.071 (1.336)
Constant	-0.453*** (-4.503)	-0.239*** (-6.540)	-0.835 (-1.279)	-0.415* (-1.702)	-0.610 (-0.783)	-0.429 (-1.500)
Controls	No	No	No	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes	Yes
Ind FE	No	No	No	No	Yes	Yes
Observations	324	324	323	323	294	294
Adjusted R2	0.041	0.030	0.054	0.029	0.057	0.042

Table 5. Risk Disclosure and Crash Risk: Information Asymmetry Channel

This table reports the regression results assessing information asymmetry on the relationship between risk disclosure and stock price crashes. *InfoAsy* is one of the following three measures: *ANADISP*, *SEG*, and *BAS*. *ANADISP* indicates analysts' earnings forecast dispersion for the firm. *SEG* indicates the number of business segments, and *BAS* indicates the average daily bid-ask spread over the last year. The dependent variable is *NCSKEW* in the odd columns and *DUVOL* in the even columns. All continuous variables are winsorized at the 1st and 99th percentiles. Controls are similar to Table 2. The constant term is not reported for brevity. *t*-statistics based on robust standard errors clustered by firm are reported in the parentheses below the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. Variables are defined in Appendix A.1.

	<i>ANADISP</i>		<i>SEG</i>		<i>BAS</i>	
	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>
	(1)	(2)	(3)	(4)	(5)	(6)
$RFD_{t-1} \times InfoAsy_{t-1}$	-0.319*** (-3.596)	-0.086*** (-2.738)	-0.007* (-1.648)	-0.002 (-1.439)	-0.504** (-2.344)	-0.215*** (-2.893)
RFD_{t-1}	-0.009 (-1.139)	-0.002 (-0.645)	-0.002 (-0.179)	0.001 (0.370)	-0.010 (-1.169)	0.000 (0.016)
$InfoAsy_{t-1}$	1.555*** (4.412)	0.437*** (3.527)	0.009 (0.535)	0.003 (0.490)	-1.254** (-2.120)	-0.254 (-1.257)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	34,944	34,944	34,637	34,637	34,803	34,803
Adjusted R ²	0.052	0.066	0.052	0.066	0.052	0.066

Table 6. Risk Disclosure and Crash Risk: Shareholder Power as a Channel

This table reports the results of tests on the potential impact of shareholder power as a channel for the impact of RFDs on stock price crash risk. Proxies for the impact of shareholder power (*Power*) include litigation risk (*LitRisk*), corporate governance measured by management entrenchment index (*E-index*; Bebchuk, Cohen, and Ferrell 2009) and institutional ownership (*IO*). For ease of interpretation, the *E-index* used in the regression is multiplied by -1 to the original value. The constant term is not reported for brevity. Control variables are the same as those in Table 2. *t*-statistics based on robust standard errors clustered by firm are reported in the parentheses below the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. Variables are defined in Appendix A.1.

	<i>LitRisk</i>		<i>E-index</i>		<i>IO</i>	
	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>	<i>NCSKEW</i>	<i>DUVOL</i>
	(1)	(2)	(3)	(4)	(5)	(6)
$RFD_{t-1} \times Power_{t-1}$	-0.011* (-1.81)	-0.004* (-1.74)	-0.014* (-1.68)	-0.008** (-2.54)	-0.072*** (-3.46)	-0.016** (-2.17)
RFD_{t-1}	-0.051** (-2.55)	-0.015** (-2.07)	-0.029 (-1.15)	-0.015 (-1.59)	0.013 (1.06)	0.004 (0.89)
$Power_{t-1}$	0.117*** (4.44)	0.043*** (4.48)	0.047 (1.51)	0.027** (2.32)	0.484*** (6.71)	0.168*** (6.55)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,200	23,200	8,426	8,426	28,808	28,808
Adjusted R ²	0.039	0.038	0.017	0.028	0.044	0.061

Table 7. Risk Disclosure and Crash Risk: Universal Demand Laws and Business Combination Laws

Panel A presents an analysis of the effect of Universal Demand (UD) laws on the relation between RFD and crash risk. The dependent variable is *NCSKEW* in column (1) and *DUVOL* in column (2). The table is similar to Table 2 except that it includes *UD*, a dummy variable equal to one in the years after firm *i*'s state of incorporation has passed a UD law and zero otherwise. Panel B presents an analysis of the effect of business combination (BC) laws on the relation between RFD and crash risk. The dependent variable is *NCSKEW* in column (1) and *DUVOL* in column (2). The table is similar to Table 2 except that it includes *BC*, a dummy variable equal to one in the years after firm *i*'s state of incorporation has passed a BC law, and zero otherwise. The year when the business combination law in each state was passed is obtained from Cain, McKeon, Solomon (2017). *t*-statistics based on robust standard errors clustered by firm and year are reported in the parentheses below the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

<i>Panel A: Universal Demand Laws</i>		
	<i>NCSKEW</i>	<i>DUVOL</i>
VARIABLES	(1)	(2)
$RFD_{t-1} \times UD$	0.030 (1.20)	0.014* (1.95)
RFD_{t-1}	-0.026** (-2.32)	-0.008* (-2.11)
<i>UD</i>	-0.288** (-2.57)	-0.106*** (-3.29)
Controls	Yes	Yes
Year FE	Yes	Yes
Industry FE	Yes	Yes
Observations	32,086	32,086
Adjusted R ²	0.099	0.088

<i>Panel B: Business Combination Laws</i>		
	<i>NCSKEW</i>	<i>DUVOL</i>
VARIABLES	(1)	(2)
$RFD_{t-1} \times BC$	0.078* (1.895)	0.027* (1.797)
RFD_{t-1}	-0.020*** (-2.749)	-0.004 (-1.457)
<i>BC</i>	-0.268* (-1.765)	-0.086 (-1.608)
Controls	Yes	Yes
Year FE	Yes	Yes
Industry FE	Yes	Yes
Observations	34,951	34,951
Adjusted R ²	0.048	0.063

Appendix A.1: Variable Definitions

Variable	Definition
<i>Accm</i>	Moving sum of the absolute value of annual discretionary accruals over the prior 3 years.
<i>ANADISP</i>	Analysts' earnings forecast dispersion measured by the standard deviation of the earnings forecasts for the firm, scaled by price of the firm at financial year-end.
<i>AT</i>	Total assets (AT) in millions.
<i>BAS</i>	The average daily bid-ask spread over the last year for a firm's stock.
<i>Beta</i>	<i>Beta</i> measured over the financial year using daily data. Firms must be traded on the NASDAQ, AMEX, or NYSE, and have traded for at least 125 trading days over the firm's financial year.
<i>BigN</i>	An indicator variable if a firm uses a Big 4/5 auditor.
<i>BM</i>	Book equity (CEQ) to market equity ($PRCC_F \times CSHO$) ratio.
<i>CRASH_DUMMY</i>	Indicator variable that takes the value of 1 if a firm experiences one or more crash weeks in a fiscal year, and 0 otherwise. We define a firm's crash weeks as those during which the firm-specific return is below 3.2 standard deviations of the average firm-specific weekly returns over the entire fiscal year.
<i>DA</i>	Debt (DLTT + DLC) as a percentage of total assets (AT).
<i>DNI</i>	Net income before extraordinary items scaled by market capitalization ($t - 1$).
<i>Dturn</i>	The average monthly share turnover over the current fiscal year minus the average monthly share turnover over the previous fiscal year, where monthly share turnover is calculated as the monthly trading volume divided by the total number of shares outstanding during the month.
<i>DUVOL</i>	The natural log of the ratio of standard deviation of down to up weeks as calculated in Eq. (3).
<i>E-index</i>	The entrenchment index from Bebchuk, Cohen, and Ferrell (2009)
<i>FileSize</i>	The file size in megabytes of the 10-K filings. We take natural logarithm of FileSize in the regression models.
<i>IO</i>	Number of shares owned by institutional investors scaled by total number of shares outstanding (winsorized to 100% maximum).
<i>IVOL</i>	Idiosyncratic volatility measured over the financial year using daily data and the Fama-French 3 factors. We annualized IVOL by multiplying by the square root of the number of trading days. Firms must have > 0 shares, be traded on the NASDAQ, AMEX, or NYSE, have common shares, and have traded for at least 125 trading days over the firm's financial year
<i>Lev</i>	Ratio of long-term debt (DLTT) over the book value of total assets (AT).

Variable	Definition
<i>LitRisk</i>	Following Kim and Skinner (2012), litigation risk is defined as $LitRisk_t = -7.883 + 0.566 \times FPS_t + 0.518 \times LNASSETS_{t-1} + 0.982 \times SALESGR_{t-1} + 0.379 \times RET_{t-1} - 0.108 \times RETSKE_{t-1} + 25.635 \times RETVOL_{t-1} + 7 \times 10^{-8} \times TURNOVER_{t-1}$, where <i>FPS</i> is an indicator variable for litigious industries that takes the value of one for firms in the biotech (SIC codes 2833–2836 or 8731–8734), computer (SIC codes 3570–3577 or 7370–7374), electronics (SIC codes 3600–3674), or retail (SIC codes 5200–5961) industry and zero otherwise (Francis et al., 1994); <i>LNASSETS</i> is the natural logarithm of total assets; <i>SALESGR</i> is the change in sales scaled by lagged total assets; <i>RET</i> is the 12-month stock returns adjusted by CRSP value-weighted market returns; <i>RETSKEW</i> is the skewness of 12-month stock returns; <i>RETVOL</i> is the standard deviation of 12-month stock returns; and <i>TURNOVER</i> is the trading volume accumulated over the fiscal year scaled by shares outstanding at the beginning of the fiscal year.
<i>Loganalyst</i>	Log of the number of analysts covering the firm at the end of the financial year. Drawn from the Thomson IBES database.
<i>Lmv</i>	The natural logarithm of market value of equity ($PRCC_F \times CSHO$) in millions.
<i>MB</i>	Ratio of the market value of equity over the book value of equity (<i>CEQ</i>). Market value of equity is the product of stock price (<i>PRCC_F</i>) and the number of shares outstanding (<i>CSHO</i>).
<i>NCSKEW</i>	The asymmetry of the return distribution by calculating the negative of the third moment of firm-specific weekly returns for each calendar year and normalizing these values by the standard deviation of firm-specific weekly returns raised to the third power. Calculated per Eq. (2).
<i>ΔREC</i>	Change in accounts receivable, defined as $REC_t - REC_{t-1}$, <i>REC</i> is accounts receivable scaled by total assets, $RECT/AT_t$.
<i>Ret</i>	The mean of firm-specific weekly returns over the fiscal year.
<i>RFD</i>	Natural logarithm of the aggregate of systematic and other risk-disclosure scores, scaled by total number of words in the 10-K filing, as measured using the word dictionary from Appendix 3 in Campbell, Chen, Dhaliwal, Lu, and Steele (2014).
<i>ROA</i>	Net income (<i>NI</i>) as a percentage of total assets (<i>AT</i>).
<i>SB</i>	Indicator variable set to 1 for firms with public equity float less than \$75 million, or revenues below \$50 million without a calculable public equity float, and 0 otherwise.
<i>SEG</i>	The number of business segments for a firm.
<i>Sigma</i>	The standard deviation of firm-specific weekly returns over the fiscal year.
<i>TURN</i>	Mean daily share turnover over the past year.